6.5 DIFFERENTIAL PULSE CODE MODULAION (DPCM)

- In analog messages (like speech, music, TV signals), the successive samples are likely to have similar amplitudes.
- Therefore, the difference between successive samples is almost small and it is mostly smaller than the maximum sample values of m(t) which is ($\pm m_p$).
- So, better performance can be achieved by transmitting the difference between the successive sample values instead of transmitting the sample values.
- Therefore, instead of transmitting m[k], we transmit the difference $d[k] = m[k] - m[k-1]$ where m[k] is the k_{th} sample of m(t) and $m[k-1]$ is the previous sample.
- Notice that the first difference sent by the encoder is the value of the first sample because the there isn't any previous sample yet, that is $m[k-1] = 0$. That is,

$d[1] = m[1] – m[0] = m[1] – 0 = m[1]$

$d[2] = m[2] – m[1]$, $d[3] = m[3] – m[2]$, and so on

- At the receiver, if d[k] and the previous sample value m[k–1] are known, we can reconstruct m[k] iteratively as follows:

$$m[k] = d[k] + m[k – 1]$$

So, $m[1] = d[1] + m[0] = d[1] + 0 = d[1]$,

$m[2] = d[2] + m[1]$, $m[3] = d[3] + m[2]$ and so on.

- In general, more efficiency is achieved if we take several previous sample values m[k – 1], m[k – 2], m[k – 3], …
- These samples constitutes the estimated value $\widehat{m}$[k].
- In this case, we transmit the prediction error d[k], which is difference between m[k] and its estimated value $\widehat{m}$[k]

$$d[k] = m[k] - \hat{m}[k]$$

- In general, the prediction error d[k] decreases as we add more samples in the past.
- The N_{th} order $\widehat{m}[k]$ is calculated as follows:

$$\hat{m}[k] = a_1 m[k-1] + a_2 m[k-2] + \cdots + a_N m[k-N] \qquad (6.44)$$

- The prediction coefficients a_j are determined from the statistical correlation between various samples. Fig. 6. 27 shows the N_{th} order predictor. The output of this predictor is $\widehat{m}[k]$, which is the predicted value of m[k] . The input consists of the previous samples m[k – 1] , m[k – 2], . . . m[k – N].
- Note that if a_1 = 1, and a_2 to a_N = 0, Eq. 6.44 reduces to first order predictor $\widehat{m}[k] = m[k-1]$. This means that the first-order predictor is a simple time delay

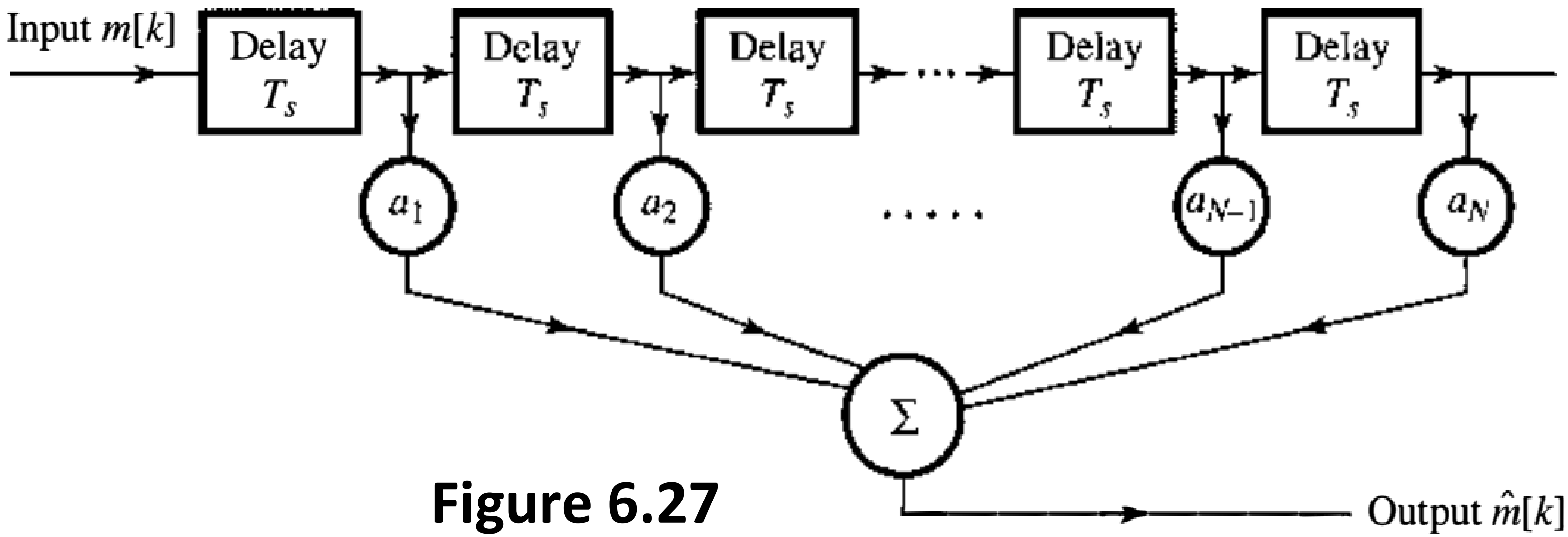


Figure 6.27

- At the receiver we also calculate the estimate $\widehat{m}$[k] from the previous sample values, and then generate m[k] as follows:

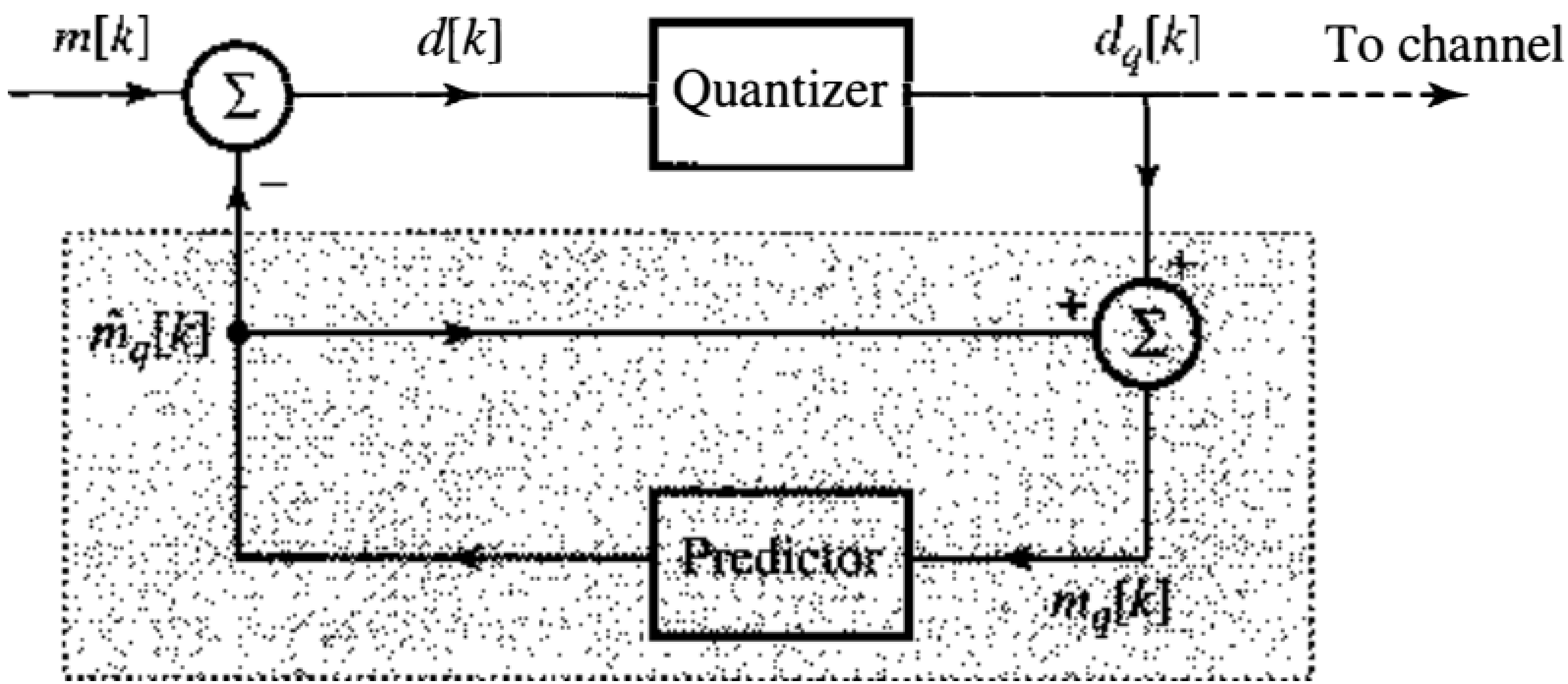
$$m[k] = d[k] + \widehat{m}[k].$$

- The first case is a special case of DPCM, where the estimate of a sample value is taken as the previous sample value, that is,

$$\widehat{m}[k] = m[k-1].$$

➢ Practical DPCM Circuit

Figure 6.28a shows a DPCM transmitter. Practical DPCM deals with quantized sample values $m_q[k]$ and sends the quantized prediction error $d_q[k]$.



(a)

Figure 6.28 DPCM system: (a) transmitter;

- From Fig.6.28, we have

$$d[k] = m[k] - \hat{m}_q[k] \qquad (6.45)$$

- The quantized prediction error is

$$d_q[k] = d[k] + q[k] \qquad (6.46)$$

where q[k] is the quantization error.

- The predictor output $\widehat{m}_q[k]$ is fed back to its input so that the predictor input $m_q[k]$ is $m_q[k] = \hat{m}_q[k] + d_q[k]$

- From Eq. (6.45), $\widehat{m}_q[k] = m[k] - d[k]$,

So, $m_q[k] = m[k] - d[k] + d_q[k]$

From Eq. (6.46), $q[k] = -d[k] + d_q[k]$

So, $m_q[k] = m[k] + q[k]$ (6,47)

- This Eq. shows that $m_q[k]$ is a quantized version of m[k].

- The receiver shown in Fig. 6.28b is identical to the shaded portion of the transmitter.

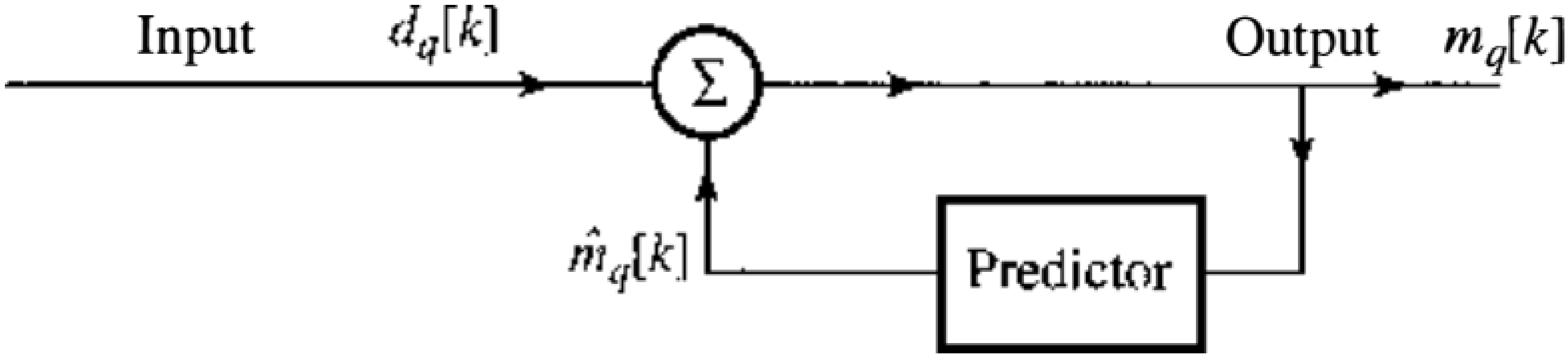


Figure 6.28 DPCM system: (b) receiver;

- The inputs in both cases are also the same, namely, $d_q[k]$. Therefore, the predictor output must be $\widehat{m}_q[k]$ (the same as the predictor output at the transmitter). Hence, the receiver output (which is the predictor input) is also the same, i.e., $m_q[k] = m[k] + q[k]$, as found in Eq. (6.47).

➢SNR Improvement of DPCM over PCM

Let m_p and d_p be the peak amplitudes of m(t) and d(t) respectively, and let Δv_P and Δv_D be the quantization steps of PCM and DPCM respectively.

Assume that we use the same value of L in both cases.

So $\Delta v_P = 2m_p/L$ and $\Delta v_D = 2d_p/L$

Hence $\Delta v_D / \Delta v_P = d_p/m_p$, so $\Delta v_D = d_p/m_p \, \Delta v_P$

This means that the quantization step of DPCM is reduced by the factor (d_p/m_p) with respect to PCM.

Reducing the quantization step Δv will reduce the quantization noise, which is given by $(\Delta v)^2/12$, which in turn improve the SNR. The quantization noise power for PCM is $q_P = (\Delta v_P)^2/ 12$ and for DPCM is $q_D = (\Delta v_D)^2/ 12$.

$\therefore \quad q_D = (d_p/m_p)^2(\Delta v_P)^2/12 = (d_p/m_p)^2 \, q_P$

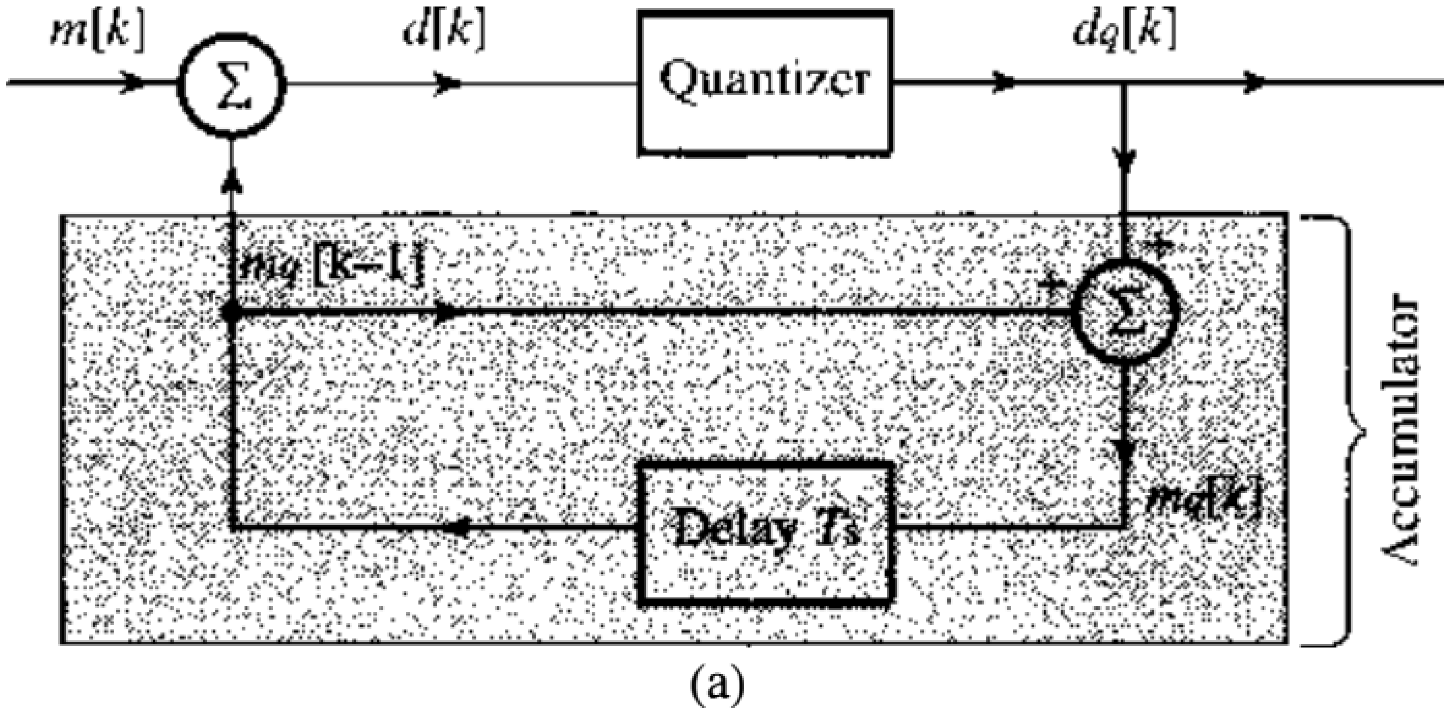
- This means that the quantization noise in DPCM is reduced by the factor $(d_p/m_p)^2$. Thus the SNR of DPCM is increased by the factor $(m_p/d_p)^2$ compared to that of PCM. Therefore, G_p (SNR improvement due to prediction) is at least $\mathbf{G_p = P_m/P_d}$ where P_m and P_d are the powers of m(t) and d(t) respectively.
- In terms of decibel units, this means that the SNR increases by $10 \log_{10}(P_m/P_d)$ dB. Therefore, Eq. (6.41) applies to DPCM also with a value of α that is higher by $10 \log_{10} (P_m/P_d)$ dB.
- In practice, the SNR improvement may be as high as 25 dB in such cases as short–term voiced speech spectra and in the spectra of low–activity images
- Alternatively, for the same SNR, the bit rate for DPCM could be lower than that for PCM by 3 to 4 bits per sample. Thus, telephone systems using DPCM can often operate at 32 or even 24 kbit/s.

6.7 DELTA MODULAION (DM)

- The correlation used in DPCM is further exploited in **DM** by oversampling (typically four times the Nyquist rate) the baseband signal.
- This increases the correlation between adjacent samples, which results in a small prediction error that can be encoded using only one bit.
- Thus DM is basically a 1–bit DPCM, that is, a DPCM that uses only two levels (L = 2) for quantization the prediction error $d[k] = m[k] - m_q[k]$.
- DM uses a first–order predictor, which is just a time delay of T_s (the sampling interval) as shown in Fig. 6.30, from which we can write

$$m_q[k] = m_q[k-1] + d_q[k] \tag{6.48}$$

Figure 6.30 Delta modulation is a special case of DPCM.



Hence $$m_q[k-1] = m_q[k-2] + d_q[k-1]$$

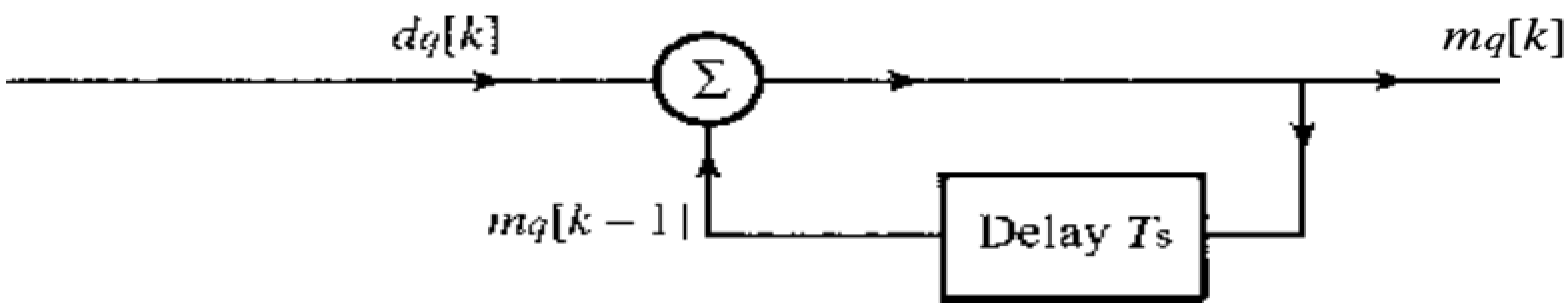
Substituting this equation into Eq. (6.48) yields

$$m_q[k] = m_q[k-2] + d_q[k] + d_q[k-1]$$

- Proceeding iteratively in this manner, and assuming zero initial condition, that is, $m_q[0] = 0$, we can write

$$m_q[k] = \sum_{m=0}^{k} d_q[m] \qquad (6.49)$$

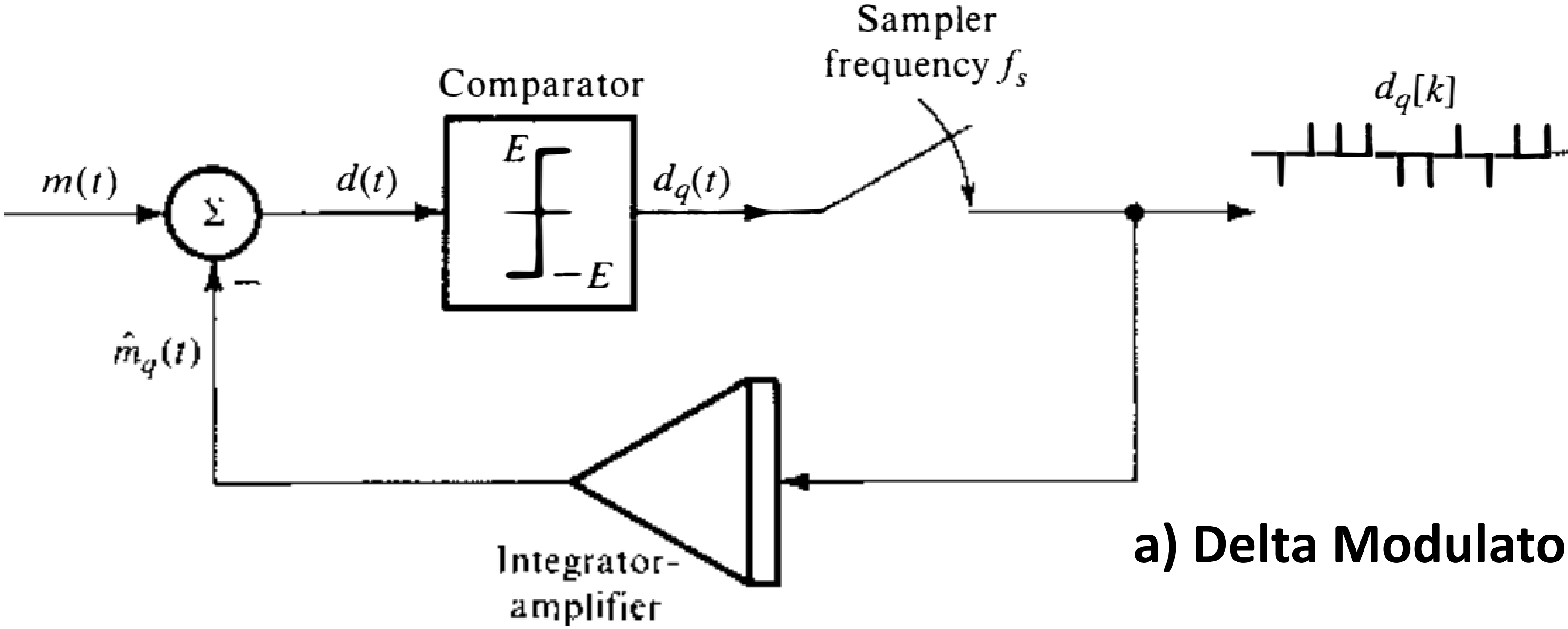
- This shows that the receiver (demodulator) is just an accumulator (adder) as depicted in figure.



(b)

- If the output $d_q[k]$ is represented by impulses, then the accumulator (receiver) may be realized by an integrator because its output is the sum of the strengths of the input impulses (sum of the areas under the impulses).
- The feedback portion of the modulator (which is identical to the demodulator) can be also replaced with an integrator.
- The demodulator output is $m_q[k]$, which when passed through a low–pass filter yields the desired signal reconstructed from the quantized samples.
- The modulator (Fig. 6.31 a) consists of a comparator and a sampler in the direct path and an integrator–amplifier in the feedback path.

- The analog signal m(t) is compared with the feedback signal (which serves as a predicted signal) $\widehat{m}_q$(t).
- The error signal d(t) = m(t) – $\widehat{m}_q$(t) is applied to a comparator. If d(t) is positive, the comparator output is a constant signal of amplitude E, and if *d*(t) is negative, the comparator output is –*E.* Thus, the difference is a binary signal (L = 2) that is needed to generate a 1–bit DPCM.
- Figure 6.31 shows a practical implementation of the delta modulator and demodulator.
- The comparator output is sampled by a sampler at a rate of f_s samples per second, where f_s is typically much higher than the Nyquist rate.



a) Delta Modulator

(a)

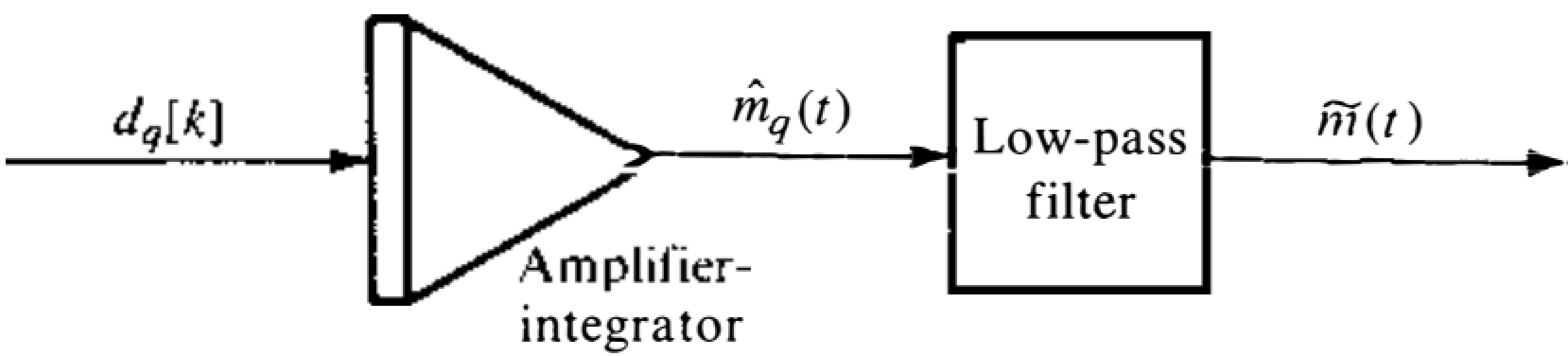
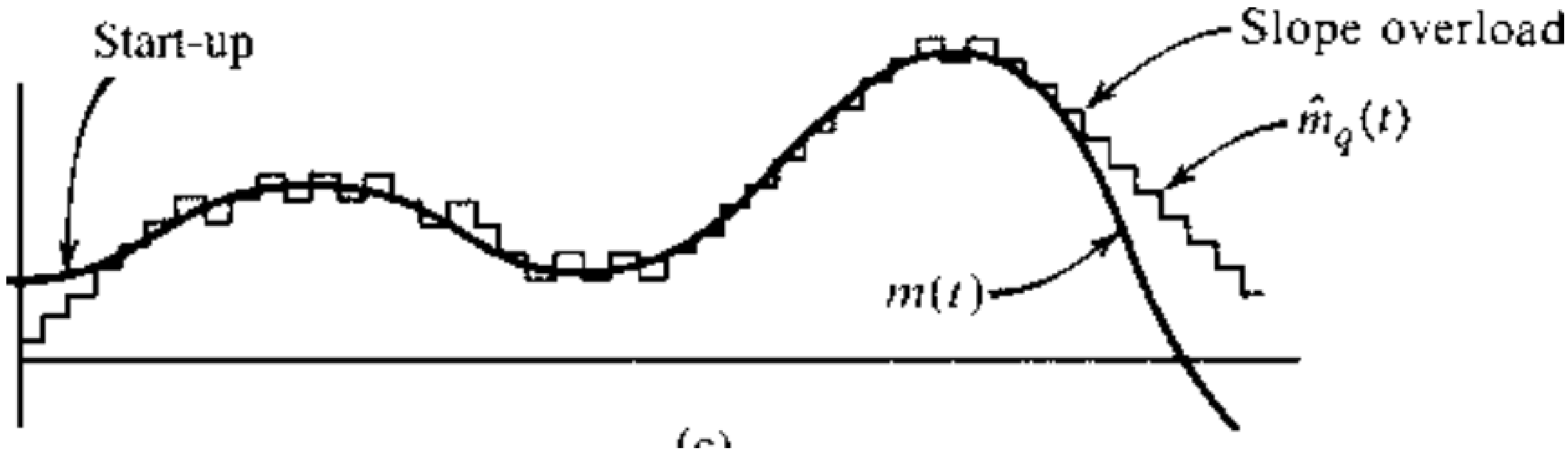


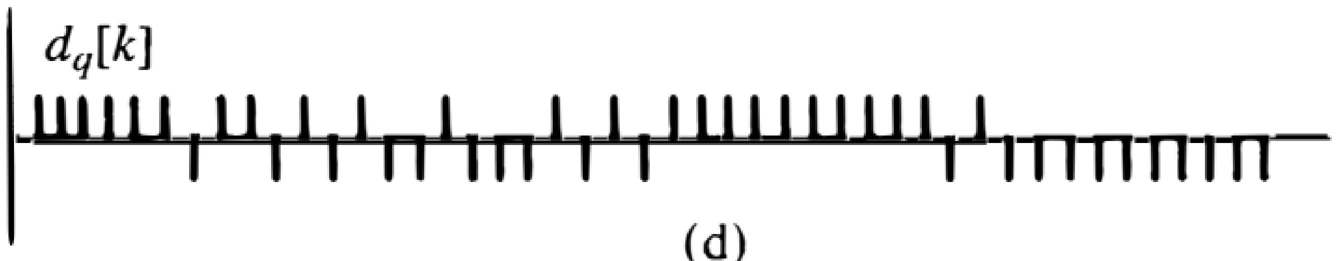
Figure 6.31

(b)

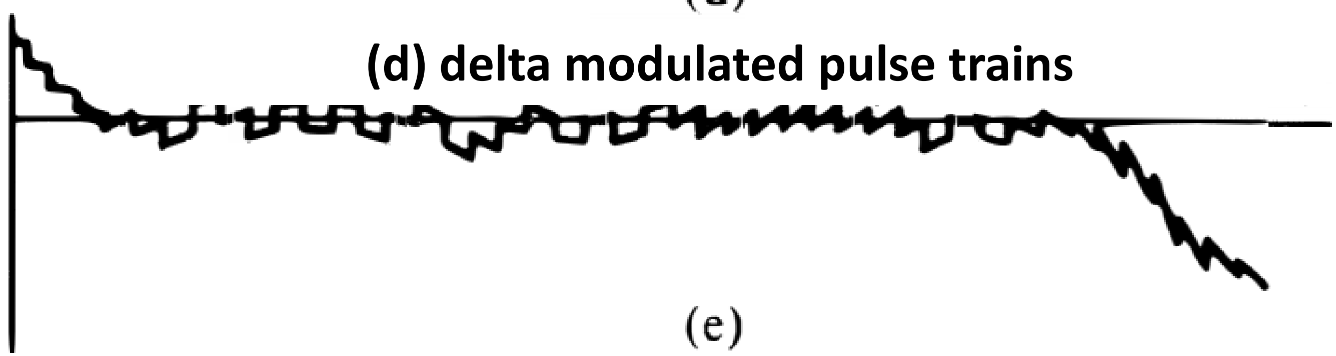
b) Delta Demodulator



c) message signal versus integrator output signal



(d) delta modulated pulse trains



(d) modulation errors

Figure 6.31

- The sampler thus produces a train of narrow pulses $d_q[k]$ with a positive pulse when $m(t) > \hat{m}_q(t)$ and a negative pulse when $m(t) < \hat{m}_q(t)$ as shown in Fig. 6.31d.
- The modulated signal $d_q[k]$ is amplified and integrated in the feedback path to generate $\hat{m}_q(t)$ (Fig. 6.3 1c), which tries to follow m(t).
- If $m(t) > \hat{m}_q(t)$, then $d_q[k] = E$. The integrator–amplifier will add E to $\hat{m}_q(t)$. This gives rise to a positive step in $\hat{m}_q(t)$, trying to equalize $\hat{m}_q(t)$ to m(t) in small steps at every sampling instant, as shown in Fig. 6.31c.
- If $m(t) < \hat{m}_q(t)$, then $d_q[k] = –E$. The integrator–amplifier will subtract E from $\hat{m}_q(t)$. This gives rise to a negative step in $\hat{m}_q(t)$, trying to equalize $\hat{m}_q(t)$ to m(t) too. It can be seen that $\hat{m}_q(t)$ is a kind of staircase approximation of m(t).

- When $\widehat{m}_q$(t) is passed through a low–pass filter, the coarseness of the staircase in $\widehat{m}_q$(t) is eliminated, and we get a smoother and better approximation to m(t).
- The demodulator at the receiver consists of an amplifier–integrator (identical to that in the feedback path of the modulator) followed by a low–pass filter (Fig. 6.31 b).
- In DM, the modulated signal is the difference between successive samples. Basically, therefore, DM sends the information about the derivative of m(t), hence, the name "delta modulation."

➢ Threshold of Coding and Overloading

There are two types of noise in DM

- **Threshold of Coding Noise** occurs when the variations in m(t) is smaller than the step value (E). This will gives rise to the granular noise in the transmitted signal similar to the quantization noise.
- **Slop–Overload Noise** occurs when m(t) changes too fast, that is if $\dot{m}$(t) is too high, then $\widehat{m}_q$(t) cannot follow m(t) and overloading occurs. This noise is one of the basic limiting factors in the performance of DM. This noise can be reduced by increasing *E* (the step size). However, this unfortunately increases the granular noise.
- There is an optimum value of E, which yields the best compromise giving the minimum overall noise.

- This optimum value of E depends on the sampling frequency f_s and the nature of the signal.
- During the sampling interval T_s, $\hat{m}_q$(t) can change by E. So, the maximum slope that $\hat{m}_q$(t) can follow is E/T_s, or Ef_s. Hence, no overload occurs if $|\dot{m}(t)| <$ Ef_s.
- Consider the case of tone modulation (meaning a sinusoidal message) m(t) = Acos ωt. The condition for no overload is

$$|\dot{m}(t)|_{\max} = \omega A < Ef_s \qquad (6.50)$$

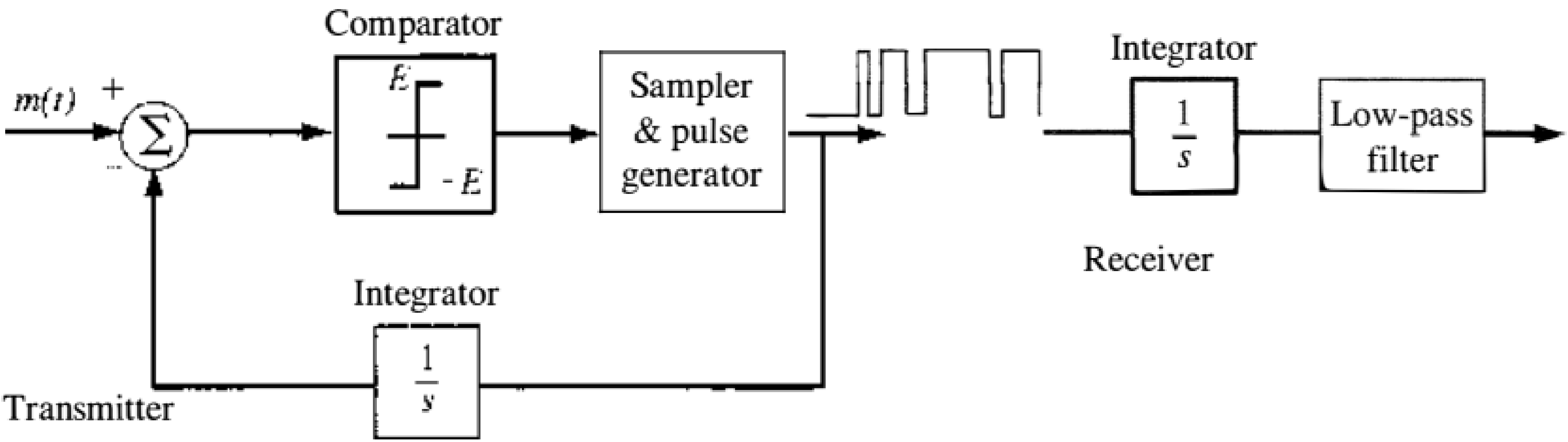
- Hence, the maximum amplitude A_{max} of this signal that can be tolerated without overload is given by

$$A_{\max} = \frac{Ef_s}{\omega} \qquad (6.51)$$

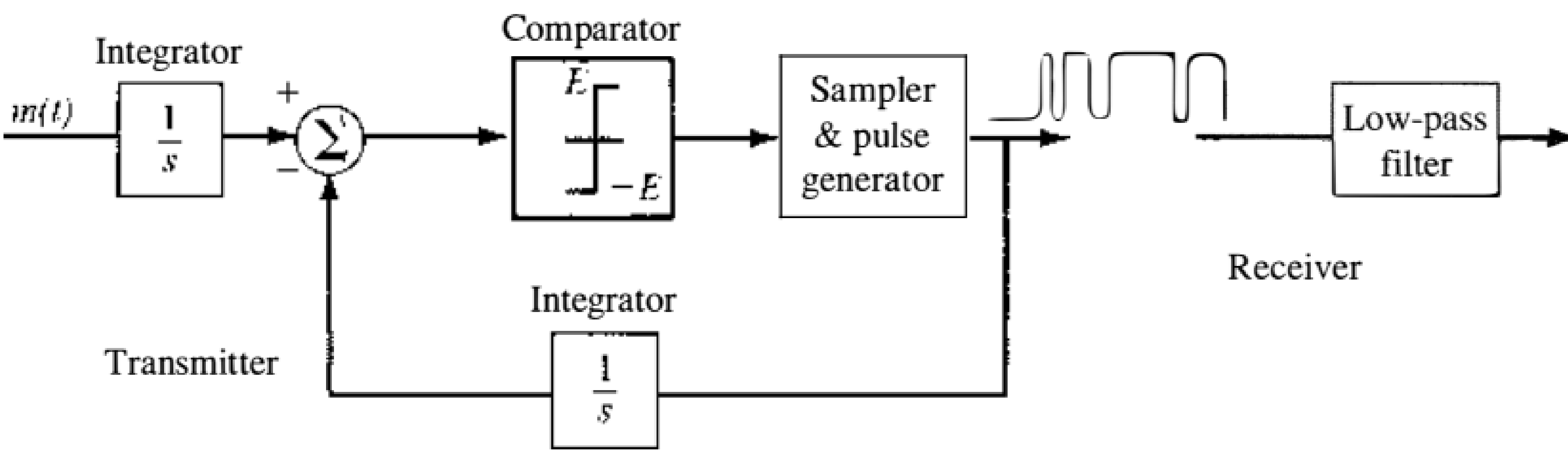
- It has been shown that A_{max} for voice signals can be calculated by using ω = 2π x 800 in Eq. (6.51).

Sigma-Delta Modulation

- The conventional DM encodes and transmits the derivative of the analog message signal.
- Hence, the receiver of DM requires an integrator as shown in Fig. 6.31 and also, equivalently, in Fig. 6.33a.
- Since signal transmission inevitably is subjected to channel noise, such noise will be integrated and will accumulate at the receiver output, which is a highly undesirable phenomenon that is a major drawback of DM.
- The sigma-delta modulation ($\sum$-Δ) overcomes this drawback. This is achieved by noting that the DM system consisting of the transmitter and the receiver as approximately distortionless and linear system.

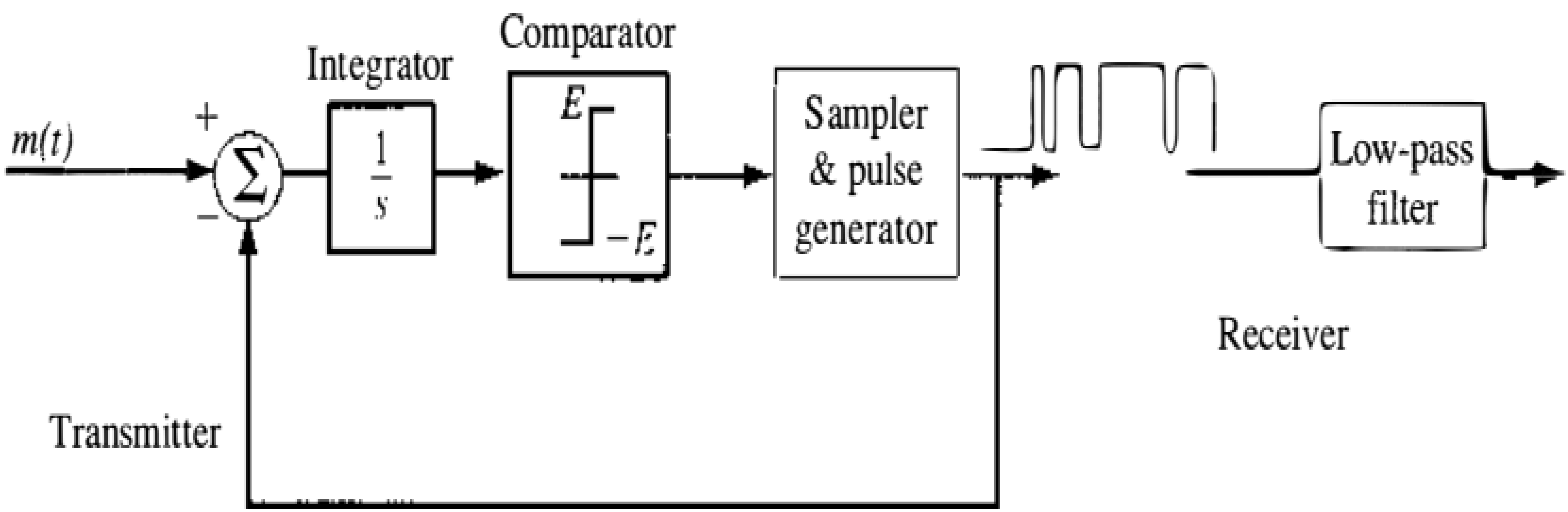

m(t)
+
Comparator
E
-E
Sampler
& pulse
generator
Integrator
1/s
Low-pass
filter
Receiver
Integrator
1/s
Transmitter

(a)


Integrator
m(t)
1/s
+
−
Comparator
E
−E
Sampler
& pulse
generator
Low-pass
filter
Receiver
Integrator
1/s
Transmitter

(b)

- Thus, the receiver integrator may be moved to the front of the transmitter (encoder) without affecting the overall modulator and demodulator response, as shown in Fig. 6.33b.
- Finally, the two integrators can be merged into a single one after the subtractor, as shown in Fig. 6.33c.



(c)